RISK AND PERFORMANCE MEASUREMENT

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Our goal is to study risk and performace of selected stocks from SP500 index and then try to create an optimal portfolio out of them. We limit ourselves to companies *KDP, KKR, KIM, KLAC, KMI, KMX, KO, KEY, KR, KMB* and timeframe from 2022-01-01 to 2025-01-01.

General statistical analysis

We begin by calculating simple returns of the stocks using the formula

$$R_i = \frac{S_i - S_{i-1}}{S_{i-1}} \tag{1}$$

where S_i is the i-th observation of the price.

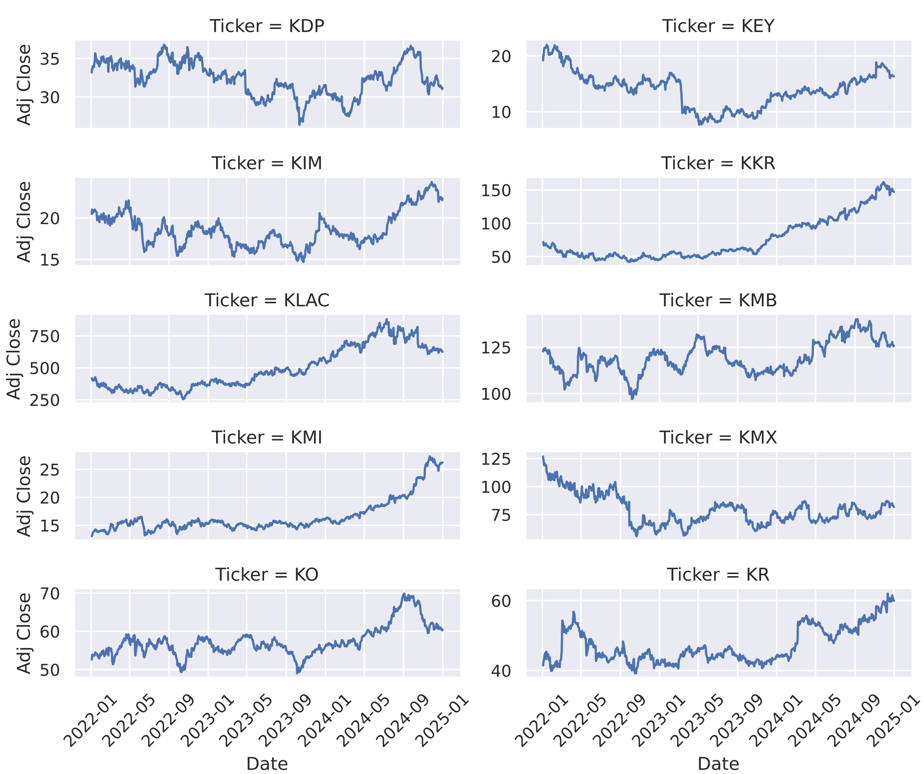

Figure 1. Adjusted close prices of the considered stocks

Ticker	KDP	KEY	KIM	KKR	KLAC
count	752.000000	752.000000	752.000000	752.000000	752.000000
mean	-0.000012	0.000163	0.000258	0.001256	0.000897
std	0.011886	0.027166	0.016898	0.024038	0.026755
min	-0.055211	-0.273308	-0.056658	-0.092944	-0.146966
25%	-0.006599	-0.013536	-0.009834	-0.012262	-0.013958
50%	0.000279	-0.000608	0.000503	0.001245	0.000297
75%	0.007301	0.012874	0.010394	0.015819	0.017049
max	0.050673	0.157010	0.075085	0.119873	0.091047

Ticker	KMB	KMI	KMX	KO	KR
count	752.000000	752.000000	752.000000	752.000000	752.000000
mean	0.000102	0.001023	-0.000213	0.000233	0.000621
std	0.011693	0.013570	0.026994	0.009839	0.016241
min	-0.057305	-0.051136	-0.246008	-0.069626	-0.073223
25%	-0.005995	-0.006413	-0.014279	-0.005277	-0.007712
50%	0.000564	0.001012	-0.000529	0.000706	0.000444
75%	0.006500	0.009045	0.013998	0.005645	0.009046
max	0.081265	0.066370	0.100741	0.038671	0.116062

Table 1. Selected statistics of the returns

From tabular data in table 1 we can observe that we can get the highest expected return from stocks *KKR* and *KMI*. *KMX, KLAC* and *KEY* have the greatest standard deviations.

When viewing histograms (fig. 3) and boxplots (fig. 4) we can notice that distribution of the returns for each stock is bell-shaped, however they are tail heavy, as indicated by the presence of the outliers in boxplots, hinting that the returns might not be normally distributed.

There are no strong correlations between the returns of individual stocks, however there are many moderate ones (between 0.4 and 0.6), such as *KO* and *KDP*, *KLAC* and *KKR* and *KO* and *KMB*. Notably, *KR* and *KLAC* are the least correlated at correlation coefficient equal to 0.034 (fig. 5)

Risk and performance analysis

Now we proceed to measure the risk and performance of the stocks. We begin with a fund of $2000 that we invest in the stocks individually. Since we are investing into stocks directly (and not, for example, their derivatives) and we assume no impact of time on the price, value of the portolio is simply the sum of the price of each stock multiplied by their amount in the portfolio. Therefore, we can calculate the P&L simply by scaling (and averaging in the case of multiple risk factors) the returns,

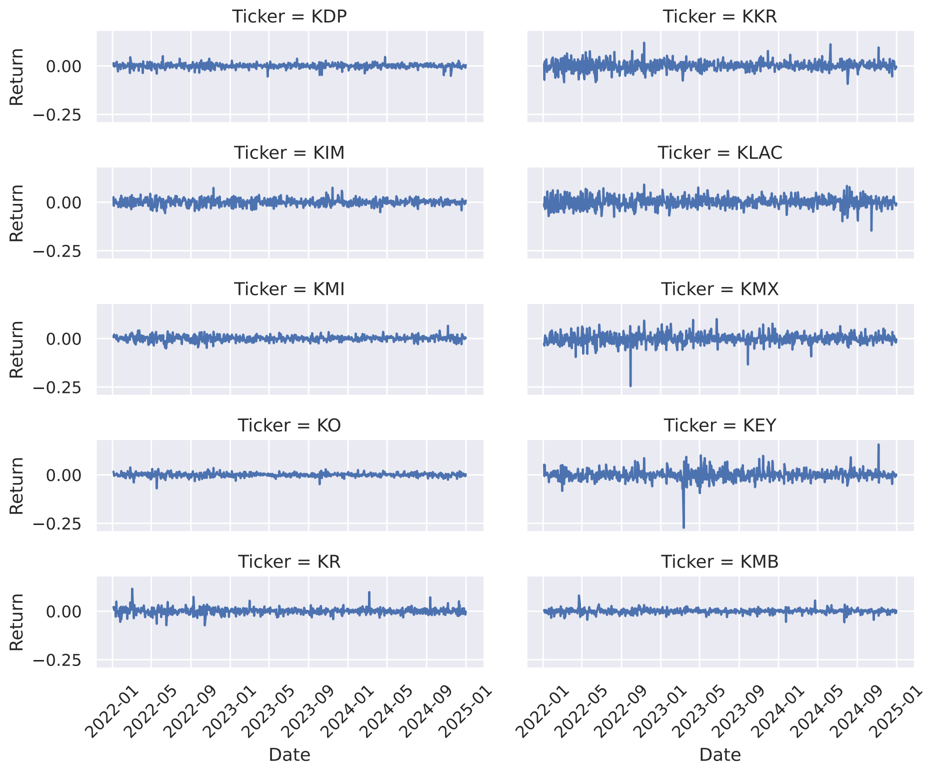

FIGURE 2. Returns of the stocks

as shown in the following derivation:

$$
\begin{aligned}
\text{P\&L}_i &= P(S_i^1, \dots, S_{i-1}^k) - P(S_{i-1}^1, \dots, S_{i-1}^k) \\
&= \sum_{j=1}^{k} a_i^j \left(S_i^j - S_{i-1}^j \right) \\
&= \sum_{j=1}^{k} a_i^j S_{i-1}^j \frac{S_i^j - S_{i-1}^j}{S_{i-1}^j} \\
&= V_i \sum_{j=1}^{k} w_i^j R_i^j
\end{aligned} \tag{2}
$$

where a_i^j is the amount of stock j in the portfolio at time i, $V_i = \sum_{j=1}^{k} a_i^j S_i^j$ is the total value of the portfolio at time i and $w_i^j = \frac{a_i^j S_i^j}{V_i}$ is the weight of the stock j in the portfolio at time i.

We measure expected value, standard deviation, value at risk, expected shortfall, Sharpe Ratio and Gain Loss Ratio for the P&Ls. Mean provides information about expected daily gain from the investment, standard deviation provides information about how far we can expected realized return from the mean. Value at risk and

FIGURE 3. Histograms of the returns

expected shortall provide information about how much money is necessary to protect the invested capital from going into negative. We also choose Sharpe Ratio and Gain Loss Ratio to measure performance of the investment. Sharpe Ratio is a simple performance measurement that considers the average gain from a portfolio and divides it by its volatility (standard deviation). Gain Loss Ratio is similar to Sharpe Ratio, however it only punishes losses instead of general volatility and it is also monotone.

From table 2 we can see that *KEY, KIM, KKR, KLAC, KMX* on average lose money and have high variance. This implies that they probably are not a good investment. Stocks with the gratest Sharpe Ratios are *KMI, KD* and *KO*. Incidentally, they also have the greatest Gain Loss Ratios. Based on that they seem to be the best investments from the considered stocks.

We also consider a portfolio optimized in the following way: for risk aversion coefficients $\gamma \in [-0.05, 0]$ we find the weights for which the mean-variance criterion is maximized. It can be interpreted that for a given mean we find weights of the portfolio that minimaze the variance, and therefore risk.

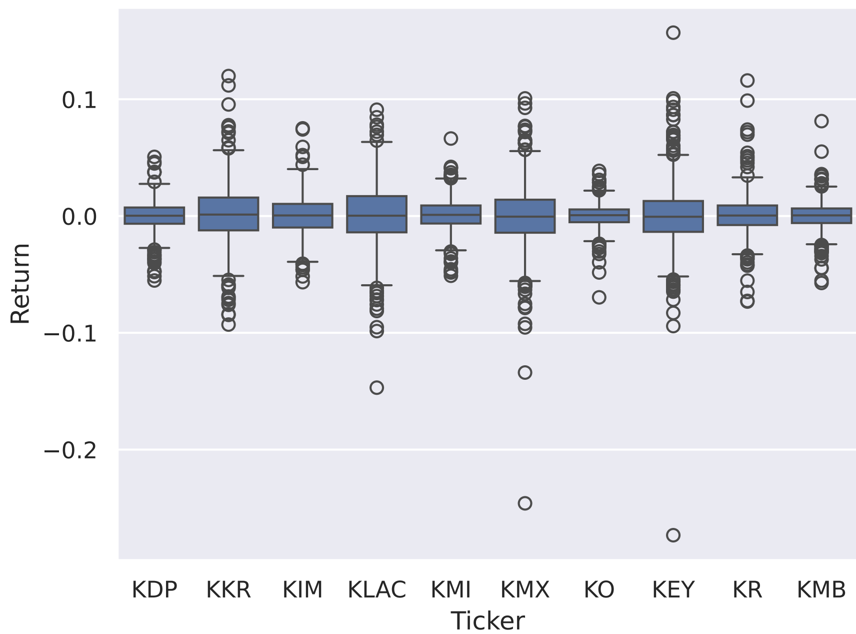

FIGURE 4. Boxplots of the returns

Ticker	KDP	KEY	KIM	KKR	KLAC
mean	0.111549	-1.612251	-0.518326	-2.660523	-0.096134
std	26.701978	44.740059	38.679135	60.940105	61.633868
VaR 1%	73.803531	106.918129	91.817132	158.588664	143.473237
ES 2.5%	69.980046	107.528323	90.282911	150.180229	132.200728
SR	0.004178	-0.036036	-0.013401	-0.043658	-0.001560
GLR	0.005100	0.000000	0.000000	0.000000	0.000000

Ticker	KMB	KMI	KMX	KO	KR
mean	0.089794	1.592800	-4.767310	0.949275	0.485229
std	27.666452	34.391119	65.107958	24.846839	42.065761
VaR 1%	71.431328	95.653861	174.067389	72.100231	137.662384
ES 2.5%	67.721324	89.683336	209.756450	71.995370	114.439372
SR	0.003246	0.046314	-0.073222	0.038205	0.011535
GLR	0.004353	0.058961	0.000000	0.048219	0.016835

TABLE 2. Performance of $2000 in each stock

We can see that, as the risk tolerance increases, all statistics rise and weights become less diversified. For the optimum portfolio we choose portfolio which maximizes Sharpe Ratio, which rougly corresponds to $\gamma = -0.002$.

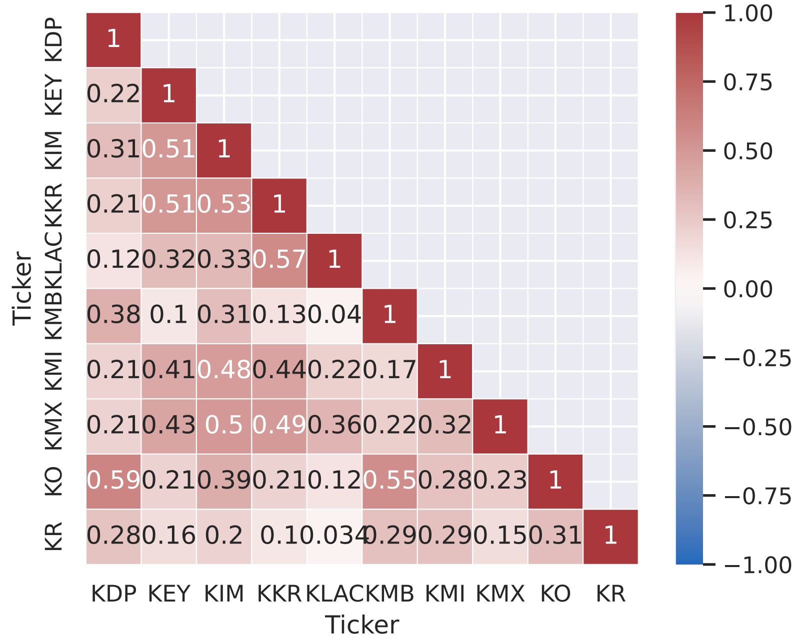

Figure 5. Correlation matrix of returns

	Chosen	Other	All	Optimized
mean	0.884541	-1.297075	-0.642590	1.291076
std	22.426515	35.170556	29.687338	24.861026
VaR 1%	58.018137	97.583469	79.200238	64.934100
ES 2.5%	56.840641	90.297836	77.390130	65.397958
SR	0.039442	-0.036880	-0.021645	0.051932
GLR	0.048367	0.000000	0.000000	0.065753

Table 3. Performance of mixed portfolios of $2000

From tables table 2 and table 3 we can see that the optimized portfolio has greater Sharpe Ratio and Gain Loss Ratio than any individual stock. Portfolio of best individual stocks with equal weights has lower performace than the best individual stock, but at much lower volatility, VaR and ES. Portfolio of other stocks and all stocks don't perform well, but they have relatively small standard deviations. Diversification has greatly reduced standard deviation, and in the case of the optimized portfolio, has increased performance of individual stocks.

In my opinion, the best investment is the optimized portolio, because it only has slightly greater variance, VaR and ES than the mix of the 3 best performing portfolios, while having much greater mean, SR and GLR.

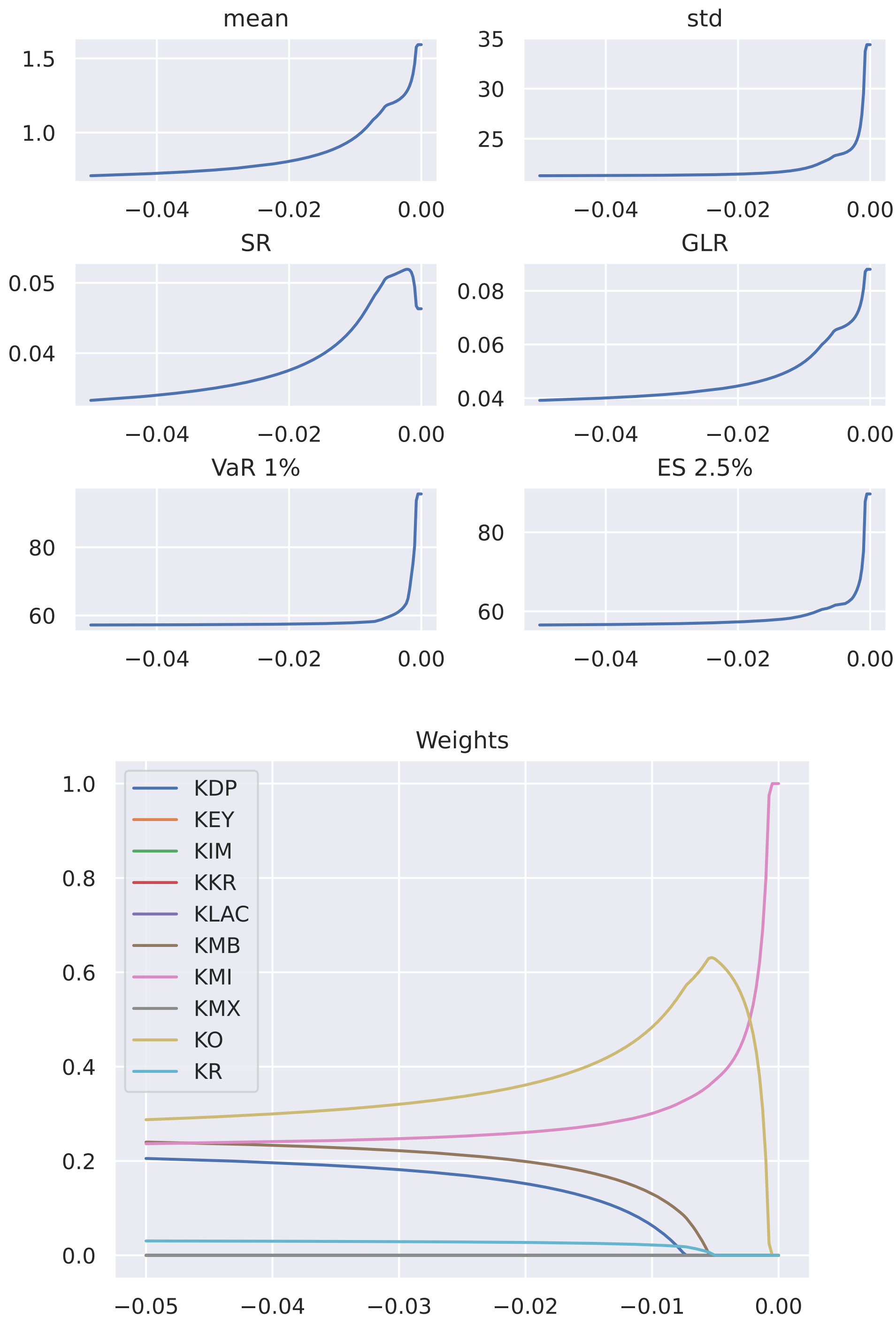

FIGURE 6. Statistics and weights for different values of γ

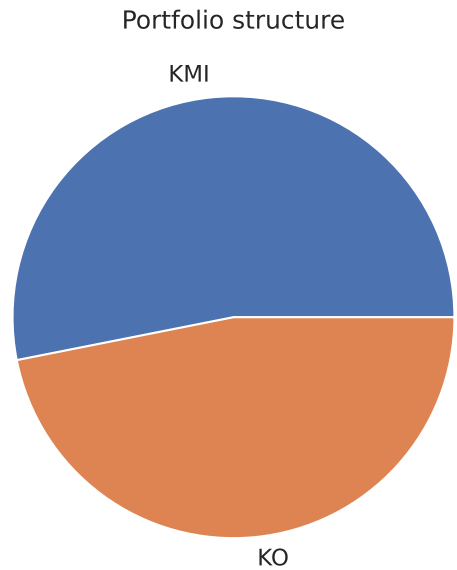

FIGURE 7. Structure of optimal portfolio. It contains only 2 stocks

Portfolio	statistic	p value
All	0.076	0.1
Optimized	0.076	0.1

TABLE 4. Results of PIT portfolio goodness of fit test to uniform distribution

Backtesting

In order to test our hypotheses, we perform a VaR backtest on our portfolios. As the consequence of eq. (2), we only need to backtest the returns.

As we can see in fig. 8, we have experienced no exceptions, meaning that our predictions have successfully protected our portfolios. However, this doesn't mean that our prediction correctly models price movement. In order to measure that, we perform PIT backtest, that is, check if the empirical distribution of the returns correctly model realized returns.

Visual inspection of fig. 9 suggests that both average portfolio and optimized portfolio pass the backtest, as the resulting distribution is close to uniform. It is further confirmed by the Kolmogorov-Smirnov test, as the results from table 4 can't refute the hypothesis, that the distribution is uniform.

The performance of the portfolios in 2024 is presented in table 5. We can see that the optimal portfolio still has greater mean, SR and GLR than portfolio of

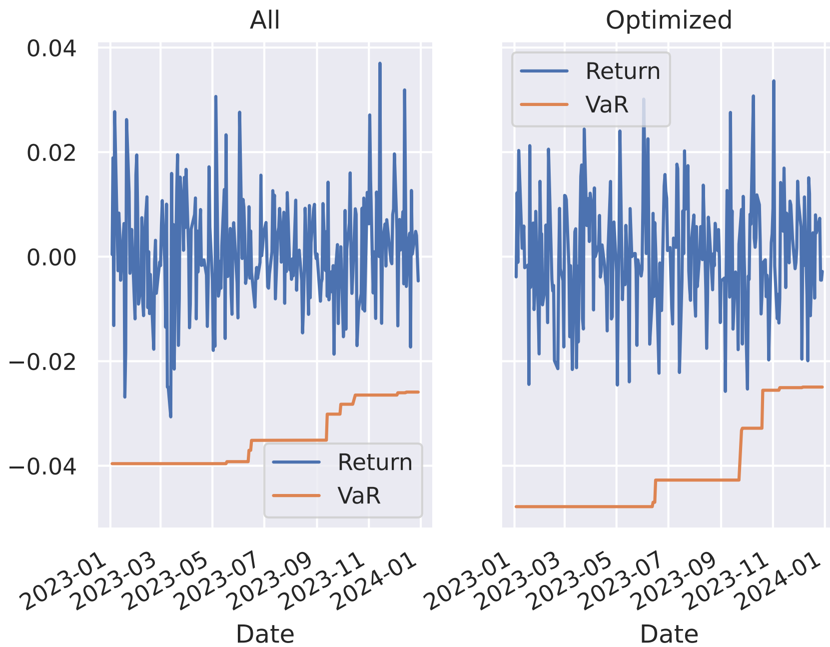

FIGURE 8. The results of the VaR backtest on the portfolios

	All	Optimized
mean	1.950528	4.087038
std	16.695499	23.444081
VaR 1%	46.097739	58.313855
ES 2.5%	41.987442	52.764781
SR	0.116830	0.174331
GLR	0.163592	0.280288

TABLE 5. Performance of portfolios in 2024-2025

all stocks. However, its std, VaR and ES are also greater. Because of that, for an investor with higher tolerance for risk, the optimized portfolio seems like a better investment. There is a moderate correlation between the portfolios, as the correlation coefficient is equal to 0.57.

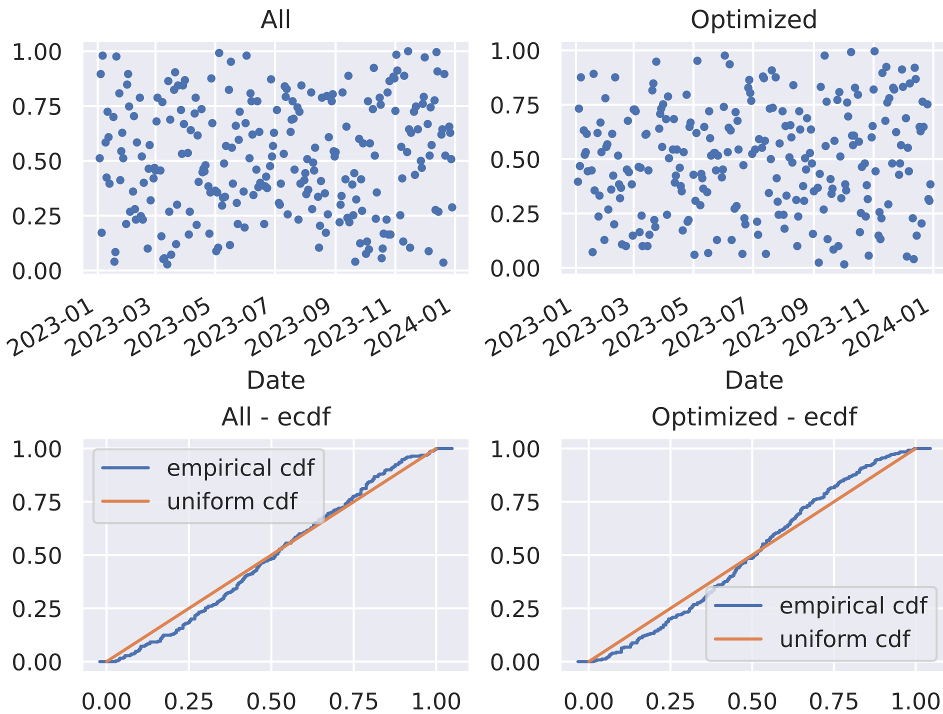

FIGURE 9. The results of the PIT backtest on the portfolios